Indian Institute of Space Science and Technology

Department of Mathematics

Maths Club — Weekly Challenge #6

Date: November 17, 2025

Challenge Question — Difficulty: ★★★★★

Let $f(x) = x^3 - \frac{3}{2}x^2 + x + \frac{1}{4}$. For every $n \in \mathbb{N}$ let f^n denote f composed n-times, i.e.,

$$f^n(x) = \underbrace{f \circ f \circ \cdots \circ f(x)}_{n \text{ times}}.$$

Evaluate $\int_0^1 f^{2025}(x)dx$.

Instructions:

- Submit your detailed solution (typed or handwritten) by **28 Nov, 2025**.
- Use the link below for submission:

Submit Here: Maths Club – Weekly Challenge Submission

- Selected solutions will be featured in the **Maths Club GitHub repository**.

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For queries: `mathsclub@iist.ac.in`